EXPLICIT DESCRIPTION OF THE CLOSURE POWERS OF AN OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11179-11189, following the statement at 11165-11177.

Editorial note. No omissions: the outer equalities are executed from named lemmas plus induction; the middle equality is a matter of notation.

Allowed standard prerequisites (editorial).

- Associativity and smd-invariance of the star operation. lemma-associativity-star and lemma-smd-star (same chapter, cited at ~11179). Reduce the left equality to an induction.
- Operations commute with unions. remark-operations-unions (same chapter, cited at ~11184). Passes the union over m through smd and star.
- Cone description of closure powers. lemma-cone-n (same chapter, cited at ~11188). Corresponding statement gives the final equality.

Source reference. Original lemma 0FX1: sources/stacks-derived/pinned-original-source.tex, source lines 10942–10942. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FX4: sources/stacks-derived/pinned-original-source.tex, source lines 10992–10992. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FX2: sources/stacks-derived/pinned-original-source.tex, source lines 10956–10956. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0FX6: sources/stacks-derived/pinned-original-source.tex, source lines 11035–11035. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{T}$ be a triangulated category. Let E be an object of $\mathcal{T}$. For $n \geq 1$ we have*

$$\langle E \rangle_n = smd(\langle E \rangle_1 \star \ldots \star \langle E \rangle_1) = smd(\langle E \rangle_1{}^{\star n}) = \bigcup\nolimits_{m \geq 1} smd(add(E[-m, m])^{\star n})$$

For $n, n' \geq 1$ we have $\langle E \rangle_{n+n'} = smd(\langle E \rangle_n \star \langle E \rangle_{n'})$.

Proof. The left equality in the displayed formula follows from Lemmas 0FX1 and 0FX2 and induction. The middle equality is a matter of notation. Since $\langle E \rangle_1 = smd(add(E[-\infty, \infty])])$ and since $E[-\infty, \infty] = \bigcup_{m \geq 1} E[-m, m]$ we see from Remark 0FX6 and Lemma 0FX2 that we get the equality on the right. Then the final statement follows from the remark and the corresponding statement of Lemma 0FX4. □